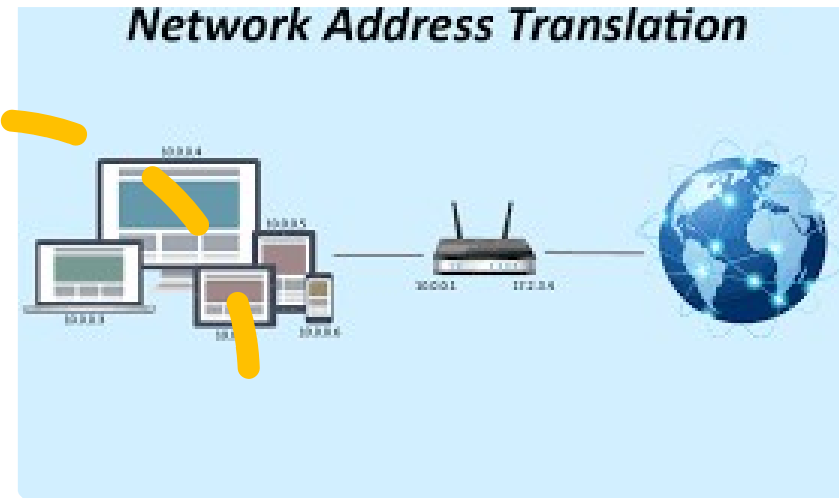


Network Layer
Network Address
Translation, IPV6

Network Address Translation (NAT).

- NAT enables a user to have a large set of addresses internally and one address, or a small set of addresses, externally.
- The traffic inside can use the large set; the traffic outside, the small set.



- To separate the addresses used inside the home or business and the ones used for the Internet, the Internet authorities have reserved **three sets of addresses** as **private addresses** shown below.

Addresses for private networks

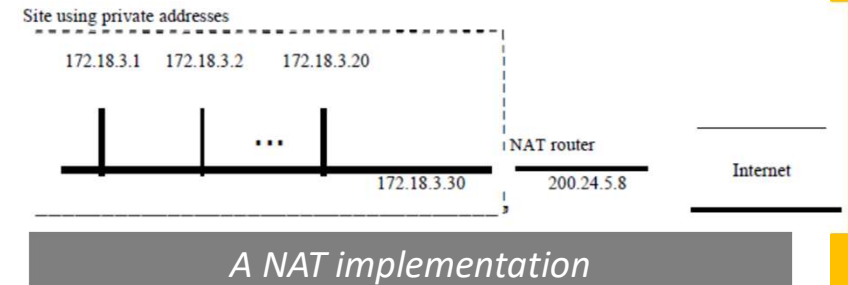
<i>Range</i>			<i>Total</i>
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

- No router will forward a packet that has one of these addresses as the destination address.



NAT implementation and Address Translation

- A private network uses private addresses. The router that connects the network to the global address uses one private address and one global address.
- The private network is transparent to the rest of the Internet; the rest of the Internet sees only the NAT router with the address 200.24.5.8.



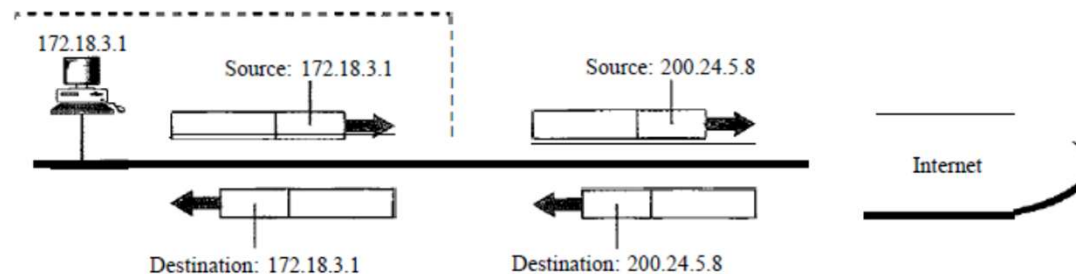
NAT implementation and Address Translation



All the outgoing packets go through the NAT router, which replaces the source address in the packet with the global NAT address.

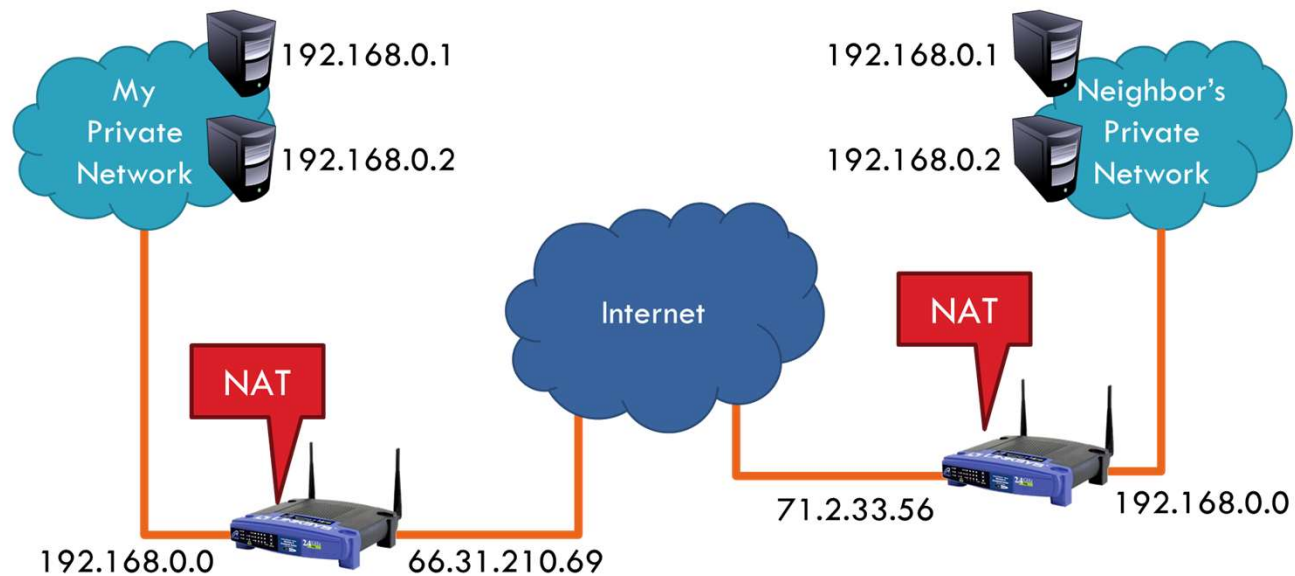


NAT router also replaces the destination address in the packet (the NAT router global address) with the appropriate private address in all incoming packets



Addresses in a NAT

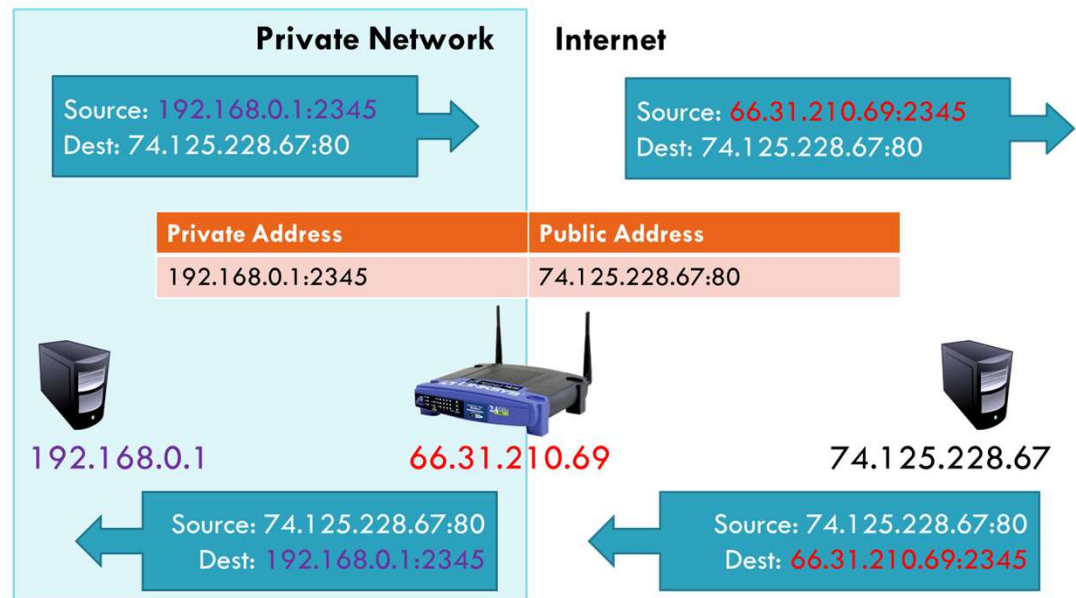
NAT IMPLEMENTATION



NAT IMPLEMENTATION

- NAT allows hosts on a private network to communicate with the Internet
 - ▣ Warning: connectivity is not seamless
- Special router at the boundary of a private network
 - ▣ Replaces internal IPs with external IP by modifying packet headers
 - This is “Network Address Translation”
 - ▣ May also replace TCP/UDP port numbers
- Maintains a table of active flows
 - ▣ Outgoing packets initialize a table entry
 - ▣ Incoming packets are rewritten based on the table

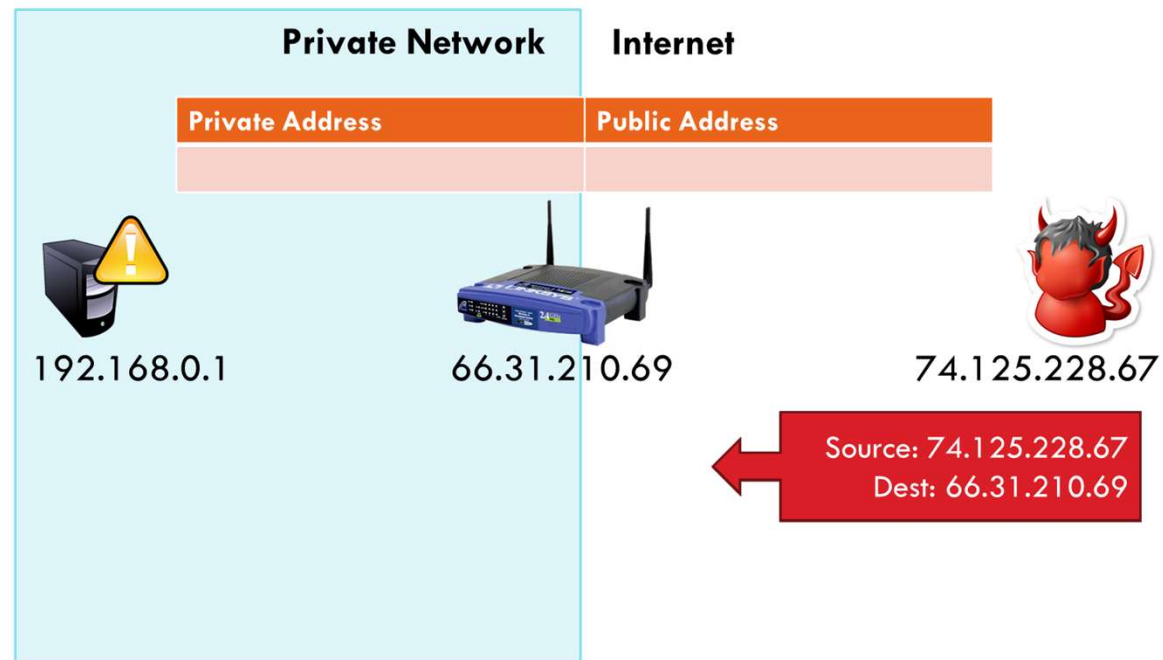
Basic NAT Operation



NAT Advantages

- Allow multiple hosts to share a single public IP
- Allow migration between ISPs
- Even if the public IP address changes, you don't need to reconfigure the machines on the LAN
- Load balancing
- Forward traffic from a single public IP to multiple private hosts

Natural Firewall



Concerns About NAT

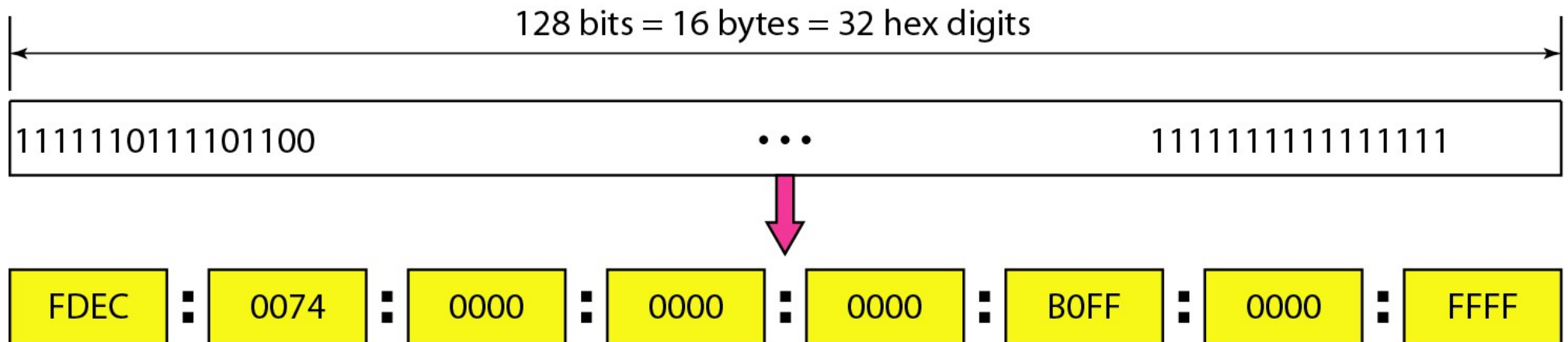
- ❑ Performance/scalability issues
 - ▣ Per flow state!
 - ▣ Modifying IP and Port numbers means NAT must recompute IP and TCP checksums
- ❑ Breaks the layered network abstraction
- ❑ Breaks end-to-end Internet connectivity
 - ▣ 192.168.*.* addresses are private
 - ▣ Cannot be routed to on the Internet
 - ▣ Problem is worse when **both** hosts are behind NATs
- ❑ What about IPs embedded in data payloads?

IPv6 ADDRESSES

- *Despite all short-term solutions, address depletion is still a long-term problem for the Internet. This and other problems in the IP protocol itself have been the motivation for IPv6.*
- *An IPv6 address is 128 bits long.*



IPv6 address in binary and hexadecimal colon notation



Abbreviated IPv6 addresses

Original

FDEC :: 0074 :: 0000 :: 0000 :: 0000 :: B0FF :: 0000 :: FFF0



Abbreviated

FDEC :: 74 :: 0 :: 0 :: 0 :: B0FF :: 0 :: FFF0



More abbreviated

FDEC :: 74 :: B0FF :: 0 :: FFF0



IPv6: Advantages



Larger address space.



Better header format.



New options.



Allowance for extensions.

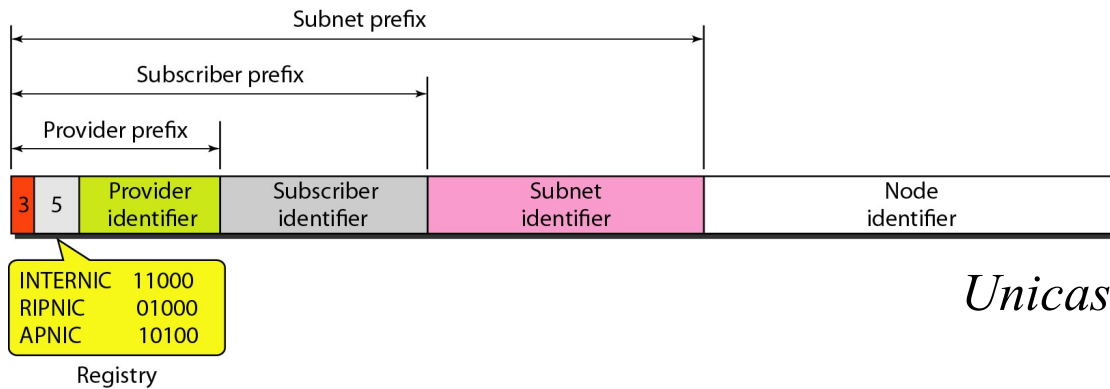


Support for resource allocation.



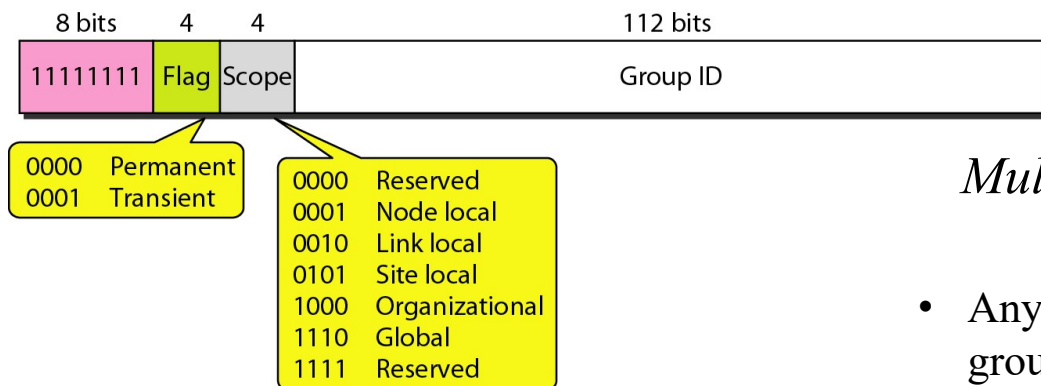
Support for more security.

Unicast and Multicast address in IPV6



Unicast address in IPv6

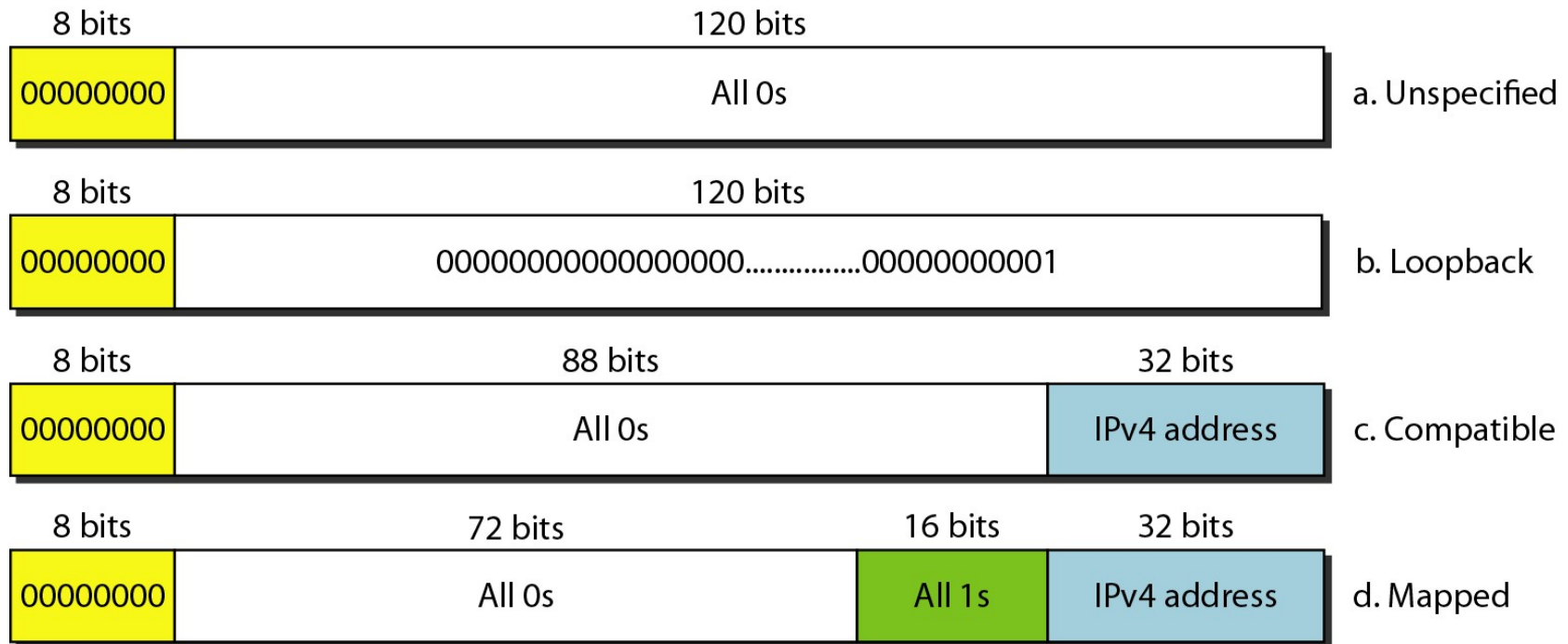
- A **unicast address** defines a single computer.
- Multicast addresses are used to define a group of hosts instead of just one.



Multicast address in IPv6

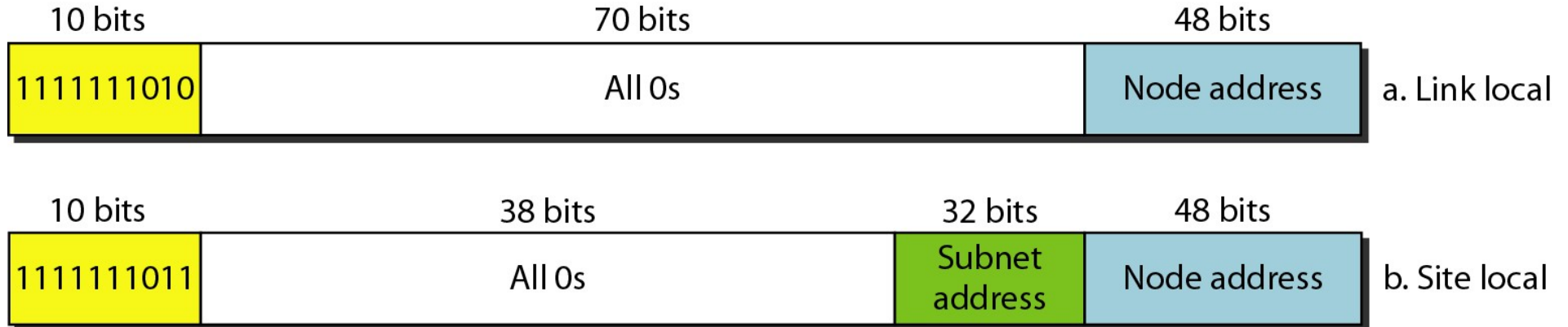
- Anycast address, like a multicast address, also defines a group of nodes.

Reserved addresses in IPv6



Local addresses in IPv6

- These addresses are used when an organization wants to use IPv6 protocol without being connected to the global Internet. In other words, they provide addressing for private networks.
- A link local address is used in an **isolated subnet**; a site local address is used in an **isolated site with several subnets**.



Question

- Which command will show you all the translations active on your router?

A. show ip nat translations

B. show ip nat statistics

C. debug ip nat

D. clear ip nat translations

Question

Expand the address 0:15::1:12:1213 to its original.

SOLUTION

XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX

0: 15:

1: 12:1213



0000:0015:0000:0000:0000:0001:0012:1213